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Clinical Sports Medicine Update

Rehabilitation and Return to Play Protocols After Anterior Cruciate Ligament Reconstruction in Soccer Players

A Systematic Review

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Background: Rehabilitation after anterior cruciate ligament ACL reconstruction (ACLR) is crucial for safe return to play (RTP) and reducing the chances of a reinjury. Yet, there is no consensus on the ideal functional tests to assess rehabilitation progress in soccer players after ACLR.

Purpose: The primary objective was to highlight the existing gap in the literature concerning the most effective standardized rehabilitation protocols and testing for facilitating successful RTP among soccer players.

Study Design: Systematic review; Level of evidence, 4.

Methods: A systematic review using PubMed, Embase, and the Cochrane Central Register of Controlled Trials (CENTRAL) was conducted. Inclusion criteria encompassed original studies (level of evidence 1-4) that examined rehabilitation protocols, metrics of knee rehabilitation, and clinical outcomes after ACLR in soccer players.

Results: This review incorporated 23 studies, predominantly retrospective case series, with a total number of 874 soccer players who underwent ACLR and rehabiliation. 5 (21.7%) studies utilized an accelerated rehabilitation protocol, while 7 (30.4%) of studies utilized a criterion-based rehabilitation. A wide heterogeneity of data was extracted including functional tests of rehabilitation and RTP such as strength test batteries, hop test batteries, and movement quality assessments. Of the 23 selected studies, 2 (8.7%) used all 3 test batteries, 8 (34.8%) used 2 test batteries, 12 (52.2%) used 1 test battery, and 1 (4.3%) used 0 of the test batteries. The mean time between surgery and RTP ranged from 3 to 8 months with only 2 (8.7%) studies reporting complications after ACLR. Lastly, out of the total studies examined, 9 (39.1%) assessed patient-reported outcome measures (PROMs), all of which demonstrated significant improvement from the initial assessment to the final follow up.

Conclusion: Soccer-specific rehabilitation after ACLR lacks standardization. Even though many studies have assessed protocols for optimal RTP and reduced secondary ACL injuries, there is a gap in the literature regarding the most effective protocols and RTP testing. The methodology reported by Kyritsis et al could serve as a foundation for future prospective randomized multicenter studies to establish a standard rehabilitation protocol and enable a successful return to soccer.

Keywords: soccer; football; ACL reconstruction; rehabilitation; return to sport; return to play

The anterior cruciate ligament (ACL), the primary stabilizer of the knee, is divided into 2 bundles. The anteromedial bundle acts to restrain anterior tibial translation, while the posterolateral bundle provides rotational stability.[4,27,52] Athletics involving jumping, pivoting, and rapid changes in direction inherently have a high risk of ACL injuries.[4,25,31,53] Soccer is the most popular sport worldwide and one in which sustaining an ACL injury can significantly hinder an athlete's performance and professional career.[7,14,41] Previous literature has evaluated different surgical approaches, tunnel preparations, and types of grafts to provide the best prognosis postoperatively to return to play (RTP). ACL reconstruction (ACLR) is determined to be the standard surgical treatment method and provides the best prognosis for athletes to return to play (RTP).[31,32,35] However, there is a lack of consensus regarding optimal rehabilitation and RTP testing after ACLR in soccer players to minimize the risk of reinjuries.[#]

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[#]References 5, 8, 16, 31, 35, 52, 53, 58, 59.

Early and progressive rehabilitation after ACLR is essential to the recovery process in athletes and begins within the first week postoperatively.[4,10,31,46,52,58] Standardized rehabilitation is usually divided into 4 phases: (1) protection and controlled ambulation, (2) controlled training, (3) intensive training, and (4) RTP.[7,8,15,31] Some studies emphasize criterion-based rehabilitation, focusing on individualized progression through protocols based on each soccer player's unique outcomes and activity.[4,15,25,30,31,34,55] In contrast, other protocols aim to accelerate the recovery process by implementing aggressive programs that plan for RTP in 6 months rather than the conservative 12 months.[7,26,46,52,55,58] In addition to various rehabilitation protocols, professional opinions often differ regarding when a player is cleared to RTP.[8,20,47,48] Therefore, the high variability of rehabilitation protocols; RTP testing; and expectations between players, physical therapists, surgeons, athletic trainers, and coaches may contribute to inconsistent rates of returning to baseline activity and soccer performance.[4,8,47,53]

Delaying RTP for 9 months in athletes engaged in pivoting sports has been found to reduce the risk of reinjuries.[41] However, a recent study by Forsythe et al[17] reported that, on average, professional soccer players in leagues of the Union of European Football Associations returned to play at 7.1 months after ACLR, which was found to be considerably shorter than other professional leagues: National Hockey League (8.5 months), National Football League (12.2 months), Major League Baseball (13.7 months), and National Basketball Association (13.9 months). Because of the unique demands of each sport, a systematic review by Warner et al[56] advocated the need to identify sport-specific factors to optimize long-term outcomes in athletes treated with ACLR. Given the significantly shorter RTP time in professional soccer players than what is recommended in the literature, there is a need to enhance outcomes that can potentially lead to long-term success.

Therefore, the main objective of this systematic review was to aggregate and evaluate all studies reporting rehabilitation in soccer players to address the existing gap in the literature concerning the most effective standardized rehabilitation protocols and testing for successful RTP among soccer players. In addition, a critical analysis of the literature reporting rehabilitation in soccer players was conducted to identify any deficits and provide a future direction for research, with the goal of implementing an evidence-based RTP functional protocol to optimize outcomes after ACLR.

METHODS

Search Strategy

A comprehensive search of the literature reporting rehabilitation protocols or functional measures of rehabilitation in soccer players who have undergone ACLR was performed in July 2022 by a reviewer (I.S.H.). Databases that were searched included PubMed, Embase, and the Cochrane Central Register of Controlled Trials (CENTRAL). Study selection was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines by 2 independent reviewers (M.A.M. and I.S.H.).[33] The following search terms were used for each aforementioned database: ("ACL" OR "anterior cruciate ligament") AND "Reconstruction" AND ("soccer" OR "football") AND "Rehabilitation." Inclusion criteria consisted of original studies, level of evidence 1 to 4, studies focusing on rehabilitation protocols and/or rehabilitation-focused functional tests of the knee after ACLR in soccer players, and studies reporting clinical outcomes after rehabilitation. The exclusion criteria consisted of studies concentrating on American or Australian football and rugby players, review articles, technique articles, conference/presentation abstracts, case reports, editorials, expert opinions, non–English-language studies, data pertaining to sports that were not exclusively soccer or stratified to report soccer-specific data, studies reporting on ACLR combined with other ligamentous procedures, and studies that did not include rehabilitation protocols or functional measures of rehabilitation. After a final selection of studies based on these criteria, we employed cross-referencing to enhance the comprehensiveness of our review. We thoroughly examined the reference lists of all 23 articles included in our analysis. This search yielded no additional articles that met the specific criteria for inclusion in our review.

Risk-of-Bias Assessment

A single independent researcher (M.A.M.) evaluated the risk of bias of all nonrandomized studies (level of evidence

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2-4) using the Methodological Index for Non-Randomized Studies (MINORS).[49] The MINORS assessment includes 8 criteria for noncomparative studies and an additional 4 criteria for comparative studies. Comparative study designs involve comparing multiple groups or conditions to assess differences, while noncomparative study designs focus on exploring and understanding a single group or condition without comparisons. Each study was given a score of 0 (not reported), 1 (reported but inadequate), or 2 (reported and adequate) for each criterion. Noncomparative studies can achieve a maximum global score of 16, whereas comparative studies have the potential for a higher maximum global score of 24. The global scores for noncomparative and comparative studies were reported as mean ± standard deviation. A higher MINORS score corresponds to a higher quality study and methodology with a lower risk of bias.

Version 2 of the Cochrane risk-of-bias tool for randomized trials (RoB 2)[50] was utilized to evaluate the risk of bias for 2 randomized controlled trials with a level of evidence 1 included in the review.

Data Extraction

Data from studies selected for the review meeting all inclusion criteria were then extracted and input into a database created by 2 independent researchers (M.A.M. and I.S.H.). Data collected included article title, authors, country, publication year, study design, level of evidence, sample size, and patient characteristics (sex, age, height, and body mass index [BMI]). Furthermore, the type of graft used for ACLR, rehabilitation protocol and duration of the protocol, clinical outcomes after rehabilitation, complications, time from injury to surgery, time from surgery to rehabilitation, time from surgery to RTP, and follow-up time after surgery were included. Patient-reported outcome measure (PROM) scores, including those for the Tegner activity scale, the International Knee Documentation Committee (IKDC) form, the Lysholm scale, the Knee injury and Osteoarthritis Outcome Score (KOOS), the Knee Outcome Survey–Sports Activity Scale (KOS-SAS), and the Cincinnati Knee Rating System, were collected and summarized. Lastly, the primary outcomes analyzed included data from isokinetic, single hop, triple hop, crossover hop, and running tests.

Statistical Analysis

Weighted means for age, height, and BMI were determined using the database created by the authors. Additionally, the pooled percentage of male participants included in the studies and the pooled percentage of hamstring tendon and bone–patellar tendon–bone (BPTB) grafts used were also determined. Ranges were calculated using the mean time from injury to surgery, mean time from surgery to rehabilitation, mean duration of rehabilitation, mean time from surgery to RTP, and mean follow-up time after surgery for all studies reporting each respective variable.

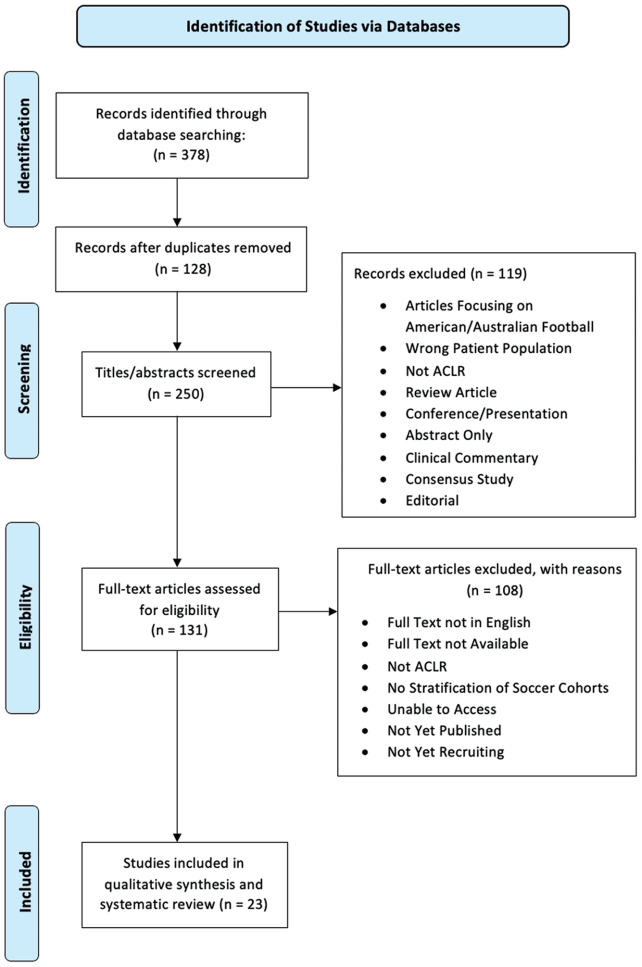

Figure 1. PRISMA flow diagram of search strategy.

RESULTS

Study Selection

In accordance with PRISMA guidelines, the database search resulted in a total of 378 articles. There were 128 duplicate articles removed, and 250 articles were eligible for the screening phase. After screening the titles and abstracts, 119 articles were excluded according to the inclusion and exclusion criteria, resulting in 131 articles eligible for full-text screening. After screening full texts, 108 studies were excluded based on the same criteria, resulting in 23 articles for a qualitative synthesis and systematic review (Figure 1).

Study Characteristics

The articles selected for this review had a level of evidence 1 to 4. Of the 23 included articles, 2 (8.7%) studies were randomized controlled trials with level 1 evidence,[32,37] 5 (21.7%) were level 2,[1-3,36,57] 8 (34.8%) were level 3,[9,23,29,40,41,45,51,54] and 8 (34.8%) were level 4.[6,12,13,21,38,42,43,59] The majority of

TABLE 1
Characteristics of Studies[a]

First Author (Year)	Level of Evidence	Study Design	Type of Study	Country
Almeida[1] (2018)	2	Cohort study	Prospective	Brazil
Angelozzi[2] (2012)	2	Single-cohort study	Prospective	Italy
Arundale[3] (2019)	2	Cohort study	Prospective	Sweden
Bisciotti[6] (2016)	4	Case series	Retrospective	Italy/Qatar
Bortone[9] (2021)	3	Cross-sectional cohort study	Retrospective	Italy
Chaves[12] (2012)	4	Cross-sectional case series	Retrospective	Brazil
Della Villa[13] (2012)	4	Case series	Retrospective	Italy/USA
Herrington[21] (2018)	4	Observational cross-sectional case series	Prospective	UK
Jacopetti[23] (2016)	3	Case-control study	Retrospective	Italy
Lee[29] (2016)	3	Case-control study	Retrospective	Republic of Korea
Mohammadi[32] (2013)	1	Randomized controlled trial	Prospective	Iran
Nambi[36] (2021)	2	Quasi-experimental study	Prospective	Saudi Arabia
Olivier[37] (2010)	1	Randomized controlled trial	Prospective	France
Ozdemir[38] (2018)	4	Case series	Retrospective	Turkey
Rahova[40] (2022)	3	Case-control study	Retrospective	Turkey
Read[41] (2020)	3	Between-participant cross-sectional study	Retrospective	Qatar/UK
Read[42] (2020)	4	Case series	Retrospective	Qatar
Read[43] (2022)	4	Repeated-measures case series	Retrospective	Qatar
Rudroff[45] (2003)	3	Cohort comparison study	Retrospective	Germany
Thomson[51] (2018)	3	Case-control study	Retrospective	Qatar/UK
van Melick[54] (2019)	3	Cross-sectional case-control study	Retrospective	Netherlands
Welling[57] (2019)	2	Case-control study	Prospective	Netherlands, Belgium, Luxembourg, Germany
Zaffagnini[59] (2014)	4	Case series	Retrospective	Italy
Total	Level 1: 2/23 (8.7%); level 2: 5/23 (21.7%); level 3: 8/23 (34.8%); level 4: 8/23 (34.8%)		Prospective: 8/23 (34.8%); retrospective: 15/23 (65.2%)	

[a]ACLR, anterior cruciate ligament reconstruction.

the studies included in this review had level 3 or 4 evidence (16/23 [69.6%]) (Table 1).

The total number of soccer players who underwent ACLR and rehabilitation was 874, with the pooled percentage of male participants being 100% (95% CI, 95%-100%; I^2 = 0%). The pooled mean age of these participants was 23.69 years (95% CI, 22.62-24.77 years; I^2 = 95%), the pooled mean height was 1.77 m (95% CI, 1.75-1.78 m; I^2 = 94%), and the pooled mean BMI was 23.57 (95% CI, 23.22-23.92; I^2 = 41%). Of the studies reporting graft data, all ACLR procedures were performed using autografts: 375 hamstring tendon, 192 BPTB, and 1 quadriceps tendon (Table 2). The pooled percentage of hamstring tendon grafts used was 63% (95% CI, 47%-78%; I^2 = 97%), while the pooled percentage of BPTB grafts used was 52% (95% CI, 35%-70%; I^2 = 99%).

Study Quality

After evaluating the selected studies using MINORS criteria, the 8 noncomparative studies included in this systematic review had a mean global score of 11.63 ± 1.19 (range, 10-14). The 13 comparative studies had a mean global score of 19.23 ± 2.68 (range, 15-23) (Appendix Table A1, available in the online version of this article). The risk-of-bias assessment for 2 level 1 evidence studies found that the studies by Mohammadi et al[32] and Olivier et al[37] corresponded to an overall "low" and "high" risk, respectively.

Rehabilitation After ACLR

A summary of the specific rehabilitation protocols (if available) is provided (Table 3). Overall, 5 (21.7%) studies subjected soccer players after ACLR to an accelerated rehabilitation protocol,[2,12,29,32,45] while 7 (30.4%) studies utilized a criterion-based/goal-targeted rehabilitation protocol.[6,36,43,51,54,57,59] The remaining 11 (47.8%) studies did not classify their rehabilitation protocols as accelerated or criterion based.**

Because of the heterogeneity of the functional measures of rehabilitation and varying tests of function related to the knee, the vast amount of data was recorded in separate tables. Although the type of test batteries utilized in each study was vastly different, the most common tests used for RTP clearance were strength test batteries, hop test batteries, and movement quality assessments. Data regarding jumping and hopping (Appendix Table A2, available online); agility and movement (Appendix Table A3, available online); knee laxity, stability, stiffness, and range of motion (Appendix Table A4, available online); and musculoskeletal strength, power, and endurance (Appendix Table A5, available online) are reported. Overall, 2 (8.7%) studies used all 3 test batteries, 8 (34.8%) used 2 test batteries, 12 (52.2%) used 1 test battery, and 1 (4.3%) used 0 of the test batteries.

**References 1, 3, 9, 13, 21, 23, 37, 38, 40-42.

TABLE 2
Characteristics of Participants[a]

First Author (Year)	No. of ACLR Cases	Male Sex, %	Age, y	Height, m	BMI	Type of Graft Used
Almeida[1] (2018)	20	100	21.594 ± 2.676[b]	1.82 ± 0.08	24 ± 2	20 HT
Angelozzi[2] (2012)	45	100	23.4 ± 4.7	1.784 ± 0.067	24.9 ± 3.3	45 HT
Arundale[3] (2019)	117	—	19.9 ± 2.5	1.679 ± 0.053	—	114 HT, 2 BPTB, 1 QT
Bisciotti[6] (2016)	80	100	24.43 ± 6.05	1.781 ± 0.042	—	35 HT, 45 BPTB
Bortone[9] (2021)	30	100	26.9 ± 5.7	—	23.68 ± 3.11	30 HT
Chaves[12] (2012)	22	100	21.77 ± 4.45	1.79 ± 0.06	23.70 ± 1.54	—
Della Villa[13] (2012)	50	88	23 ± 6	—	—	36 HT, 11 BPTB, 3 AT
Herrington[21] (2018)	15	—	22.3 ± 3.1	1.75 ± 0.10	—	10 HT, 5 BPTB
Jacopetti[23] (2016)	17	—	26.180 ± 8.967	1.797 ± 0.050	23.376 ± 2.000	17 BPTB
Lee[29] (2016)	16 (8 PT, 8 ARE)	100	PT: 18.14 ± 1.21; ARE: 18.00 ± 1.71	PT: 1.7428 ± 0.0503; ARE: 1.7385 ± 0.0546	—	—
Mohammadi[32] (2013)	42 (21 BPTB, 21 STG)	100	BPTB: 24.9 ± 2.8; STG: 25.2 ± 2.4	BPTB: 1.783 ± 0.078; STG: 1.790 ± 0.075	—	21 HT, 21 BPTB
Nambi[36] (2021)	30 (15 acute, 15 chronic)	—	Acute: 19.86 ± 4.52; chronic: 19.72 ± 2.15	Acute: 1.77 ± 0.03; chronic: 1.74 ± 0.05	Acute: 22.8 ± 1.6; chronic: 23.1 ± 1.7	—
Olivier[37] (2010)	24 (12 OLC, 12 no OLC)	100	OLC: 25.11 ± 3.41; no OLC: 23.31 ± 3.12	OLC: 1.79 ± 0.08; no OLC: 1.81 ± 0.09	—	3 HT, 21 BPTB
Ozdemir[38] (2018)	14	100	Range, 19-38	1.76 ± 0.05	—	—
Rahova[40] (2022)	20	100	25.40 ± 5.32	1.77 ± 0.59	23.47 ± 1.20	—
Read[41] (2020)	166 (42 <6 mo, 69 6-9 mo, 55 >9 mo)	100	<6 mo: 24.3 ± 4.5; 6-9 mo: 23.7 ± 6.7; >9 mo: 24.0 ± 5.4	<6 mo: 1.751 ± 0.104; 6-9 mo: 1.747 ± 0.082; >9 mo: 1.742 ± 0.074	—	—
Read[42] (2020)	20	100	24.6 ± 4.2	1.753 ± 0.102	—	5 HT, 15 BPTB
Read[43] (2022)	27	100	24.5 ± 3.9	1.745 ± 0.071	—	8 HT, 19 BPTB
Rudroff[45] (2003)	30 (15 PAT, 15 ST)	100	PAT: 32.6 ± 4.0; ST: 29.1 ± 6.7	PAT: 1.780 ± 0.014; ST: 1.779 ± 0.015	—	15 HT, 15 BPTB
Thomson[51] (2018)	16	100	26 ± 4	1.78 ± 0.06	—	9 HT, 7 BPTB
van Melick[54] (2019)	14	100	23.2 ± 3.6	—	23.4 ± 1.8	—
Welling[57] (2019)	38	100	24.2 ± 4.7	—	—	24 HT, 14 BPTB
Zaffagnini[59] (2014)	21	100	22.9 ± 5.4	—	—	—
Total	874	Range, 88-100	Range, 18-38	Range, 1.68-1.82	Range, 22.8-24.9	375 HT, 192 BPTB, 1 QT

[a]Data are shown as mean ± SD unless otherwise indicated. "—" denotes variables that were not reported. ACLR, anterior cruciate ligament reconstruction; ARE, accelerated rehabilitation exercise; AT, achilles tendon; BMI, body mass index; BPTB, bone–patellar tendon–bone; HT, hamstring tendon; OLC, one-leg cycling; PAT, patellar tendon; PT, physical therapy; QT, quadriceps tendon; ST, semitendinosus tendon; STG, semitendinosus and gracilis tendon.
[b]Value was approximated using median and interquartile range.

Timeline From Injury to RTP

The various timelines for soccer players from injury to ACLR, ACLR to rehabilitation, and ACLR to RTP are also reported (Table 4). The mean time from injury to surgery ranged from 8 days to 46 months, while the mean time from surgery to rehabilitation ranged from 2 days to 6 months. The mean length of rehabilitation ranged throughout the studies from 6 weeks to >9 months, and the mean time from surgery to testing ranged from 2 weeks to 4 years. The mean time between surgery and RTP ranged from 3 to 8 months, while the mean follow-up time after surgery ranged from 4 to 19.2 months. Only 1 (4.3%) of the studies included data on the rate of RTP. The study by Zaffagnini et al[59] found that 95% of the patients returned to the same initial professional soccer level; 1 patient was able to RTP but in a lower division.

PROM Scores

There were 9 (39.1%) studies that reported PROM scores. The Tegner activity scale was utilized in 4 (17.4%) studies, with initial scores ranging between 3.5 and 9.0 and final follow-up scores ranging between 3.9 and 9.0. Additionally, 3 (13.0%) studies used the IKDC form, with initial scores of 44.0 to 68.0 and final scores of 86.5 to 93.0. The Lysholm scale was also incorporated in 3 (13.0%) studies, with an initial score of 80.0 and a range of 88.0 to 99.2 at the final follow-up. There were 2 (8.7%) studies that reported using the KOOS in which the initial score was 58.4 and the final follow-up score ranged from 89.9 to 98.2. Lastly, 1 (4.3%) study implemented the KOS-SAS, with an initial score of 79.0 and a final score of 96.0, and 1 (4.3%) study recorded using the Cincinnati Knee Rating System, with an initial score of 89.2 and a final follow-up score of 94.8 (Appendix Table A6, available online).

Complications

Only 2 (8.7%) studies reported complications after ACLR. In the study by Zaffagnini et al,[59] 1 patient suffered a new ACL injury at 1.5 years after ACLR. Additionally, Della Villa et al[13] observed 2 patients with associated lesions or surgery and reported 3 other patients with complications not explicitly stated (Table 4).

DISCUSSION

The primary findings of this review on rehabilitation protocols and functional measures after ACLR in soccer players

TABLE 3
Rehabilitation Protocols[a]

First Author (Year)	Protocol
Almeida[1] (2018)	6 mo of standard rehabilitation
Angelozzi[2] (2012)	The goal of the accelerated rehabilitation protocol was RTP within 6 mo and the early restoration of full knee extension and weightbearing. Exercises focused on neuromuscular, strength, plyometric, and agility training. Soccer-specific tasks were also used. Rehabilitation was conducted 3 times a week. After 6 mo, an additional 20-wk training program was added, with an emphasis on improving the rate of force development and maximizing power.
Arundale[3] (2019)	Rehabilitation in which patients completed protocols for <6, 6-9, and >9 mo
Bisciotti[6] (2016)	The "goal target" protocol allowed for progression based on the achievement of a well-defined functional outcome. Tests included the following: 1. Isometric yo-yo test: Perform an isometric push with both legs. 2. Isokinetic test: Perform a quadriceps contraction and hamstring contraction at 60 deg/s. 3. Synchro plates test: Perform a countermovement jump at 90° of knee flexion. During the landing phase, stop the movement at 90° of knee flexion. 4. Modified triple leg jumping test: Perform 3 jumps in the frontal plane with the injured leg and then with the noninjured leg. 5. Modified side-step cutting test: After a run of 5 m, touch a skittle, and make a change in direction at 90°. Then run at maximum speed to a second skittle 5 m away. 6. Sprint braking test: Sprint at maximum speed for 30 m. Then sprint 30 m at 90% of maximum speed to a skittle. Repeat a second trial with a skittle at 6 m and then 4 m. 7. Illinois agility test: Follow a specific path at maximum speed. 8. Modified Illinois agility test: Follow a specific path at maximum speed with a ball.
Bortone[9] (2021)	Standardized postoperative and rehabilitation protocol
Chaves[12] (2012)	Functional rehabilitation with an accelerated protocol
Della Villa[13] (2012)	Initial phases of rehabilitation: Patients began engaging in gymnasium and pool exercises that targeted pain, swelling, range of motion, proprioception, strength, and aerobic fitness. Early soccer-specific exercises were performed. Final phases of rehabilitation: Patients attended 90-min OFR sessions 2 to 5 d/wk. The protocol consisted of 5 phases of progressive soccer-specific exercises including strength training, conditioning, and increased musculoskeletal and neuromechanical functional demands.
Herrington[21] (2018)	Rehabilitation on a full-time basis within their professional soccer club
Jacopetti[23] (2016)	Same rehabilitation protocol for all patients
Lee[29] (2016)	The accelerated rehabilitation program suggested by Shelbourne and Nitz[46] was modified and applied in 2-h sessions 5 times a week. Phase 1 was intended to relieve inflammation and edema. Phase 2 aimed to increase range of motion. Phase 3 focused on increasing muscle strength. Phase 4 sought to improve balance ability.
Mohammadi[32] (2013)	Accelerated rehabilitation protocol that focused on strengthening, balance, proprioception, stability, agility, plyometrics, and RTP training
Nambi[36] (2021)	Rehabilitation protocol that focused on strengthening the quadriceps and hamstring muscles, with specific criteria for progression
Olivier[37] (2010)	At 1-2 wk: The priority was to decrease pain and swelling and to restore full knee extension and quadriceps strength. At 2-6 wk: The plan was to continue increasing knee range of motion and weightbearing and gaining hamstring and quadriceps control. Additionally, gait and static proprioception exercises were initiated. OLC: In addition to the above exercises, the untreated leg was engaged in an OLC training program. The OLC training program consisted of 3 sessions a week for 6 wk.
Ozdemir[38] (2018)	During the rehabilitation program, patients had sessions 4 to 6 times a week. Angular velocity with exercises was decreased throughout training. The angular velocity was 240 deg/s at 1 wk, 180 to 150 deg/s at 2 wk, 120 deg/s at 3 and 4 wk, and 60 deg/s at 5 and 6 wk.
Rahova[40] (2022)	The 6-mo rehabilitation program consisted of strengthening, stretching, isometric, isotonic, balance and coordination, proprioception, and plyometric exercises. Running and cycling were also initiated.
Read[41] (2020)	Standardized rehabilitation protocol
Read[42] (2020)	Rehabilitation protocol with previous 6-m hop exercises
Read[43] (2022)	Criteria-based rehabilitation program designed by van Grinsven et al[52] combined with a sports-specific phase of reconditioning
Rudroff[45] (2003)	The rehabilitation protocol lasted 6 wk. The initial program consisted of daily passive and active treatments. After the patients achieved full range of motion, weight training and an accelerated rehabilitation program that focused on muscle coordination were started.
Thomson[51] (2018)	Criteria-based rehabilitation program
van Melick[54] (2019)	Rehabilitation sessions 2 to 3 times a week based on ACLR practice guidelines that emphasized functional goal–based criteria

(continued)

TABLE 3
(continued)

First Author (Year)	Protocol
Welling[57] (2019)	The strength training protocol was based on American College of Sports Medicine principles and rehabilitation guidelines that emphasized individualized exercises. The initial phase focused on attaining full knee extension and pain reduction and performing quadriceps activation exercises. The second phase at 10 to 14 wk was composed of muscular endurance exercises. The third phase at 12 to 14 wk aimed to improve strength and normalize leg strength symmetry. Endurance training was continued with the addition of balance exercises, running, and jump-landing techniques. The fourth phase at 14 to 16 wk aimed to improve knee extension and flexion muscle strength deficits. The same exercises and variations were repeated from the previous stages. This phase finished with OFR and RTP.
Zaffagnini[59] (2014)	The rehabilitation protocol was patient specific. Patients began with gymnasium and pool sessions to decrease pain and swelling and improve range of motion, proprioception, strength, and aerobic fitness. Early soccer-specific exercises were also used. The final phases comprised OFR. Each OFR session lasted 90 min, 2 to 5 d/wk, and was composed of 5 stages. Exercises included strength training, conditioning, and increased functional demands of musculoskeletal and neuromechanical components involved in recovery after ACLR.
Total	Accelerated rehabilitation protocol: 5/23 (21.7%); criterion-based rehabilitation protocol: 7/23 (30.4%); unclassified rehabilitation protocol: 11/23 (47.8%)

[a]ACLR, anterior cruciate ligament reconstruction; OFR, on-field rehabilitation; OLC, one-leg cycling; RTP, return to play.

were as follows: (1) a wide heterogeneity with regard to the functional measures of rehabilitation and their association with successful RTP, graft failure, reinjuries, reoperations, and PROM scores; (2) the need for future studies to evaluate and consider what functional tests and rehabilitation protocols should be used in "gold-standard" practice for soccer players; and (3) the low inclusion rate of female soccer players who were evaluated in the studies and the need for further investigations, as previous literature has reported a higher risk of secondary ACL injuries in female athletes compared with their male counterparts.

Our Appendix Tables A2 to A5 demonstrate the heterogeneity regarding functional tests of rehabilitation after ACLR, with a wide array of PROMs; jump and hop tests; movement tests; knee laxity, stability, stiffness, and range of motion tests; and musculoskeletal strength, power, and endurance tests. The lack of consistency in measurements of rehabilitation progression and the lack of studies that report RTP, graft rupture, or complication rates make it challenging to identify which functional tests of rehabilitation allow soccer players to successfully RTP. Additionally, there is a lack of long-term follow-up, with only one study, by Zaffagnini et al,[59] retesting athletes at 4 years after ACLR. Therefore, the literature on soccer players after ACLR and rehabilitation requires more standardized objective measures with adequate follow-up to elucidate functional tests with meaningful effects.

While our findings shed light on the existing body of evidence and current gaps in the literature, there are still several avenues for future rehabilitation and RTP testing that could further enhance our understanding and improve clinical practice in this area. A level 2 systematic review by Glattke et al[18] concluded that recovery protocols after ACLR remain heterogeneous in clinical practice because of the lack of consensus and variability of the quality of evidence. The authors noted that supervised rehabilitation produced better outcomes and recommended the need to design studies that focus on functional tests and RTP criteria. Additionally, the consensus article by Meredith et al[30] concluded that patients ought to follow a structured plan once certain clinical and functional goals are achieved. The decision regarding RTP should then consider various factors, such as objective physical examinations and reliable RTP testing, which incorporate assessments of functionality and have undergone validation through peer reviews. Finally, a comprehensive systematic review and multidisciplinary consensus article carried out by van Melick et al[53] proposed an evidence-based statement aimed at promoting consistency in physical therapy treatment with the application of functional performance assessments for patients after ACLR. The research group recommended the adoption of goal-oriented rehabilitation criteria encompassing isometric, concentric, and eccentric strength exercises. They also promoted the use of neuromuscular training with rehabilitation spanning a period of 9 to 12 months. Lastly, they advocated for the inclusion of rigorous test batteries for RTP clearance, comprising not only strength and hop tests but also meticulous assessments of movement quality to ensure the athlete's readiness for safe RTP.

Therefore, one approach that we propose for future work was conducted in 2016 by Kyritsis et al.[28] This key publication evaluated 158 male professional athletes, which included 105 soccer players in Qatar who underwent ACLR, criterion-based rehabilitation, and then RTP. The study aimed to analyze whether strength or functional RTP testing criteria could predict future ACL graft ruptures in male professional athletes. Before RTP, participants completed 6 functional tests, including isokinetic strength testing at 60, 180, and 300 deg/s; single hop, triple hop, and triple crossover hop tests; on-field sport-specific rehabilitation; and a running T-test. Athletes who successfully completed the tests and achieved RTP were categorized as "fully discharged" from rehabilitation, whereas

TABLE 4
Timeline From Injury to RTP[a]

First Author (Year)	Complications	Time From Injury to Surgery	Time From Surgery to Rehabilitation	Duration of Rehabilitation	Time From Surgery to Testing	Time From Surgery to RTP	Follow-up Time After Surgery
Almeida[1] (2018)	Study excluded patients with failure	4.0 ± 2.9 mo[b]	—	6 mo	6 mo	—	6 mo
Angelozzi[2] (2012)	—	4.3 ± 2.8 mo	1 wk	6 mo and additional 20 wk	6 and 12 mo	—	—
Arundale[3] (2019)	—	≤3, >3->6, 6-9, and ≥9 mo	—	<6, 6-9, and ≥9 mo	6-12, >12-24, and ≥24 mo	—	—
Bisciotti[6] (2016)	—	—	—	165.93 ± 16.73 d	Isometric yo-yo test: 90.7 ± 17.1 d; isokinetic test: 168.6 ± 12.2 d; synchro plates test: 163.6 ± 8.2 d; modified side-step cutting test: 168.6 ± 6.3 d; modified triple leg jumping test: 181.5 ± 7.2 d; sprint braking test: 165.3 ± 11.8 d; Illinois agility test: 170.6 ± 9.8 d; modified Illinois agility test: 169.6 ± 8.5 d	—	—
Bortone[9] (2021)	—	10 d	—	—	—	Already returned	Recreational: 18.8 ± 6.12 mo; competitive: 19.2 ± 2.39 mo
Chaves[12] (2012)	—	—	—	Currently in rehabilitation	—	—	4-12 mo
Della Villa[13] (2012)	5 (2 associated lesions or surgery [1 previous ACL reconstruction and 1 concomitant medial meniscal suture procedure], 3 other)	1 y	17 ± 9 d	138 ± 33 d	Every 2-4 wk during rehabilitation period	148 ± 36 d	—
Herrington[21] (2018)	—	—	—	—	—	Already returned	7.8 ± 1.3 mo
Jacopetti[23] (2016)	0	—	—	—	—	—	1 y
Lee[29] (2016)	0	—	2 d	12 wk	12 wk	—	—
Mohammadi[32] (2013)	Study excluded patients with failure	—	—	—	—	8 mo	BPTB: 8.1 ± 1.4 mo; STG: 8.2 ± 1.8 mo
Nambi[36] (2021)	—	Acute: 8-24 d; chronic: 28-46 mo	8 wk	Currently in rehabilitation	8 wk, 6 mo, and 12 mo	—	—
Olivier[37] (2010)	—	2.0 ± 0.2 mo	—	6 wk	6 wk	—	—
Ozdemir[38] (2018)	0	—	4 mo	6 wk	6 wk	—	—
Rahova[40] (2022)	0	—	6 mo	6 mo	6 mo	—	—
Read[41] (2020)	—	—	—	Currently in rehabilitation	<6, 6-9, and >9 mo	—	<6 mo: 19.5 ± 1.9 wk; 6-9 mo: 29.1 ± 3.2 wk; >9 mo: 46.0 ± 6.7 wk
Read[42] (2020)	—	—	—	—	—	—	36.0 ± 10.5 wk
Read[43] (2022)	—	—	—	Currently in rehabilitation	17 ± 5, 25 ± 6, and 34 ± 7 wk	—	—
Rudroff[45] (2003)	—	—	—	6 wk	2 y	3 mo	—
Thomson[51] (2018)	—	—	—	—	—	—	<9 and ≥9 mo
van Melick[54] (2019)	—	—	—	—	—	—	12.4 ± 3.5 mo
Welling[57] (2019)	—	—	—	38-46 wk	3.9 ± 1.11, 6.6 ± 0.72, and 9.7 ± 0.80 mo	Already returned	—
Zaffagnini[59] (2014)	1 new ACL injury at 1.5 y after ACL reconstruction	48 ± 36 d	12 ± 4 d	157 ± 49 d	3 mo, 6 mo, 12 mo, and 4 y	186 ± 52 d	—
Total		Range, 8 d to 46 mo	Range, 2 d to 6 mo	Range, 6 wk to >9 mo	Range, 2 wk to 4 y	Range, 3-8 mo	Range, 4-19.2 mo

[a]Data are shown as mean ± SD unless otherwise indicated. "—" denotes variables that were not reported. ACL, anterior cruciate ligament; BPTB, bone–patellar tendon–bone; RTP, return to play; STG, semitendinosus and gracilis tendon.

[b]Value was approximated using median and interquartile range.

athletes who failed to meet all discharge criteria and RTP were categorized as "not fully discharged." The authors found that athletes in the "not fully discharged" group had a 4 times greater risk of graft ruptures (73%) than athletes who were "fully discharged" (27%). Thus, a structured and complete set of functional tests before RTP seems to be a potential positive approach for soccer players after ACLR.

Regarding the timing of RTP, a public database study by Forsythe et al[17] found that high-level professional soccer players who were active in the Union of European Football Associations between 1999 and 2019 typically returned to sport at a mean of 7.1 months after ACLR. This is notably shorter than the 9 months recommended in the current literature for athletes participating in pivoting sports, which has been shown to decrease the rate of reinjuries by up to 51%.[19] However, Forsythe et al[17] reported that in these high-level professional soccer players, the graft rupture rates were equivalent to rates reported in other professional sports. Kyritsis et al's[28] study was similar in that it included professional soccer players who returned to sport at a mean of 7.5 months (range, 3.8-7.5 months) after ACLR. Therefore, RTP timing for clearing soccer players is also an important area of interest that requires further exploration in future studies.

Lastly, the studies included in our review demonstrated a limited representation of female soccer players after ACLR, warranting the need for additional research (Table 2). Previous literature has consistently reported a higher incidence of ACL injuries among female athletes compared with their male counterparts.[11,22,24,39,44,53] A recently published meta-analysis by Hong et al[22] assessed clinical outcomes after ACLR exclusively in soccer players and reported that female athletes had a higher likelihood of secondary ACL injuries, with an incidence rate of 27% for female athletes compared with 10% for their male counterparts. Similarly, female patients are less likely to recover from these injuries and achieve RTP at their previous levels.[11] Physiological differences in female patients may be responsible for these outcomes, such as increased knee laxity, different limb alignments, and different ligament sizes.[22,39,44,53] Therefore, further investigation focusing on female soccer players is essential to gain a comprehensive understanding of their unique rehabilitation needs and develop targeted interventions to mitigate the risk of recurrent ACL injuries.

Limitations

This review has several limitations, primarily the heterogeneity of the studies that met inclusion criteria regarding sample size, patient characteristics, rehabilitation protocols, and outcome measures after ACLR. Additionally, there was a lack of long-term follow-up data, which is critical for assessing the success of rehabilitation protocols and their efficacy in reducing graft ruptures, reinjuries, and complications. Moreover, the 2 most common grafts used for ACLR were hamstring tendon grafts and BPTB grafts. Some of the outcomes reported may be confounded by the biomechanical attributes of each respective graft. Many of the studies showed a low inclusion rate of female soccer players and were retrospective in nature. Thus, the risk of bias inherent to all retrospective studies must be taken into consideration when assessing the validity of the findings. Lastly, as this article focused on soccer players, the findings reported in this review may not necessarily translate to rehabilitation after ACLR in other sports.

CONCLUSION

There is a paucity of studies that evaluate specific rehabilitation protocols to optimize players to RTP and experience low rates of secondary ACL injuries. However, publications focusing on soccer-specific rehabilitation after ACLR currently lack standardization. Kyritsis et al's[28] methodology can be utilized as a basis for future prospective randomized multicenter studies to elucidate evidence-based RTP testing that allows soccer players to successfully return to "The Beautiful Game."

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